

Solar-Charged Emergency Light and Fan

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In scorching summer days we all feel irritated due to frequent power cuts. Emergency lights do not work long enough as their batteries are not

charged fully due to the power cuts. During night the situation is even worse as it becomes impossible to sleep when the fan also stops working.

Here is a circuit for solar-charged emergency light that can also be used to run a small DC fan. It can help in resolving the problem of uncharged batteries by providing dual charging option. Its battery gets charged by a solar panel during the day, but when it is cloudy, or at night, it can be charged using the mains power supply.

Circuit and description

The circuit is built around a 12V, 10W solar panel (connected at SP1 in Fig. 1),

step-down transformer X1, 12V-7Ah dry battery (BATT.1), 3-way rotary switch S1, 30 bright white-coloured LEDs (LED1 through LED30), 12V-1A DC fan (M1) along with some other components.

The DC fan and LED light are powered by BATT.1, which is charged by the solar panel when switch S1 is in position 2. When solar voltage is less than 12 volts, or insufficient to charge the battery—at night or on a rainy day—switch S1 can be turned to position 1 to charge the battery through mains supply. LED31 and LED32 glow to indicate that the solar voltage is sufficient for charging the battery.

This circuit uses a 12V, 10W solar panel to charge a 12V-7Ah dry battery, which is normally used in a UPS system for a personal computer. As we are using a small-capacity solar panel there is no particular requirement for a charge controller.

PARTS LIST

Semiconductors:

T1, T2, T4	- BC547 npn transistor
T3	- BC557 pnp transistor
LED1-LED30	- 5mm white bright LED
LED31-LED33	- 5mm red LED
LED34	- 5mm green LED
D1-D3	- 1N4007 rectifier diode
ZD1	- 9.1V zener diode

Resistors (all ¼-watt, 45% carbon):

R1-R8, R16, R17	- 100-ohm
R9, R12	- 2.2-kilo-ohm
R10, R13	- 10-kilo-ohm
R11, R14, R15	- 1-kilo-ohm
R18	- 1.2-kilo-ohm
VR1, VR2	- 4.7-kilo-ohm preset

Capacitors:

C1	- 470µF, 35V electrolytic
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Miscellaneous:

SP1	- 12V, 10W solar panel
X1	- 230V AC primary to 14V-0-14V, 1A secondary transformer
CON1	- 2-pin connector
CON2	- 2-pin connector, male and female pair
S1	- 3-way rotary switch
S2, S3	- On/off switch
M1	- 12V DC-operated small fan
BATT.1	- 12V, 7 Ah chargeable battery

Test Points

Test point	Details
TP0	0V, GND
TP1	16V (approx.)
TP2	Solar panel voltage (12V to 17V)
TP3	Battery voltage
LED33	Glow when battery voltage is less than preset minimum level
LED34	Glow when battery voltage is more than preset maximum level

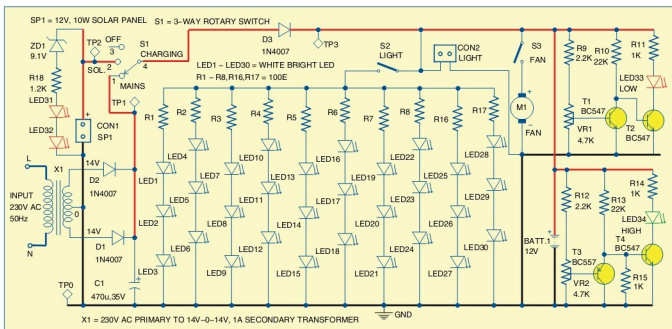


Fig. 1: Circuit for solar-charged emergency light